Template

Studentnames and studentnumbers here

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Set-up your environment

Title Page

Names: Imane Ouahi, David Abelovic, Xi Hui, Rebecca Devasia, Chaime Oh

Tutorial group number: 5 2

Tutorial lecturer's name: Simon Loop

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

The gender pay gap (GPG) remains a significant social and economic issue across the European Union (EU). Although certain progress has been made toward improved gender equality, women continue to earn less than men on average across many European countries (Eurostat, 2024). While the size of the pay gap varies between countries and regions, persistent differences in earnings remain a challenge for policymakers and labor markets across the EU Several factors contribute to the gender pay gap. This includes things such as occupational segregation, differences in working hours, career interruptions related to pregnancy and caregiving responsibilities, and unequal representation in higher-paying positions (Blau & Kahn, 2017). Recent research also highlights that structural inequalities in labour markets and workplace practices continue to influence earnings differences between men and women (International Labour Organization [ILO], 2024). Understanding the gender pay gap is important, since it affects women' s financial independence, career opportunities, and long-term economic security. Lower earnings throughout a woman's career can lead to lower pension income and an increased risk of poverty later in life (European Commission, 2024). In addition, wage inequality may also reduce economic efficiency by limiting the full utilisation of skills and talent of women within a given workforce. This study examines the gender pay gap across the EU by grouping Northern, Western, Southern, and Eastern European countries, and then comparing individual regions. Using wage, employment, and income data collected over multiple years, our research aims to identify regional patterns and trends in gender-based wage inequality. The findings may contribute to discussions on equal pay policies and gender equality for women in the European labor market.

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

midwest is an example dataset included in the tidyverse package

2.2 Provide a short summary of the dataset(s)

The gdp_data dataset:

The dataset "Gender pay gap in unadjusted form" comes from Eurostat. It measures the unadjusted gender pay gap, which is the difference in average gross hourly earnings between men and women. The dataset was last updated 3 months ago and contains 584 data cells.

- freq: Only one value is used, which is A (annual), so there is only one unique value.
- unit: Only PC (percentage) is used, so there is one unique value.
- nace_r2: Only the category [B-S_X_O] is included, so there is one unique value.
- geo: There are 39 geo codes instead of 41, meaning some geographic codes are missing from the extracted data. The missing codes seem to be: "LI" (Liechtenstein) and "EA21".
- TIME_PERIOD: The data covers 2007 to 2024.
- values: These represent the percentage difference in hourly wages between men and women.

The salary_data dataset:

The dataset "Average full time adjusted salary per employee" comes from Eurostat. This dataset shows the average salary per employee (full-time adjusted), allowing fair comparisons between countries regardless of differences in part-time work. The dataset was last updated 7 months ago and contains 1,560 data cells.

- freq: Only one value is used, A (annual), so there is one unique value.
- unit: Two units are used, EUR (euro) and NAC (national currency), so there are two unique values.
- geo: Includes 26 European countries and 2 additional aggregates, resulting in 28 unique geo entries. - TIME_PERIOD: The data covers the years from 1995 to 2024.
- values: Salaries are expressed in EUR and national currency (NAC).

```r
summary(gpg_data)
```

```
##         freq              unit             nace_r2              geo
##  Length    :584   Length    :584   Length    :584   Length    :584
##  N.unique :  1   N.unique :  1   N.unique :  1   N.unique : 39
##  N.blank  :  0   N.blank  :  0   N.blank  :  0   N.blank  :  0
##  Min.nchar:  1   Min.nchar:  2   Min.nchar:  7   Min.nchar:  2
##  Max.nchar:  1   Max.nchar:  2   Max.nchar:  7   Max.nchar:  9
##
##   TIME_PERIOD             values
##  Min.   :2007-01-01   Min.   :-1.30
##  1st Qu.:2011-01-01   1st Qu.: 9.70
##  Median :2016-01-01   Median :14.45
##  Mean   :2015-08-06   Mean   :13.87
```

```
## 3rd Qu.:2020-01-01   3rd Qu.:17.60
## Max.   :2024-01-01   Max.   :30.90
```

```r
summary(salary_data)
```

```
##        freq              unit               geo           TIME_PERIOD
## Length   :1560   Length   :1560   Length   :1560   Min.   :1995-01-01
## N.unique :   1   N.unique :   2   N.unique :  28   1st Qu.:2004-01-01
## N.blank  :   0   N.blank  :   0   N.blank  :   0   Median :2011-01-01
## Min.nchar:   1   Min.nchar:   3   Min.nchar:   2   Mean   :2010-06-16
## Max.nchar:   1   Max.nchar:   3   Max.nchar:   9   3rd Qu.:2018-01-01
##                                                    Max.   :2024-01-01
##      values
## Min.   :    964
## 1st Qu.:  14585
## Median :  25700
## Mean   :  97584
## 3rd Qu.:  38114
## Max.   :7297665
```

In this case we see 28 variables, but we miss some information on what units they are in. We also don't know anything about the year/moment in which this data has been captured.

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

2.3 Describe the type of variables included

Variables included in SALARY_DATA

- freq: This variable shows how often the data is collected over time. In this dataset, it is recorded annually (A).
- unit: The unit of measure variable shows the measurement scale used for the data values. The unit of measures used in this dataset are Euro and NAC (National currency).
- geo: The variable geological entity identifies 28 territorial units. It includes 26 individual European countries, represented by ISO 3166-1 alpha-2 codes, as well as two supranational aggregates.
- TIME_PERIOD: The time variable shows the time period covered by the dataset, which spans from 1995 to 2024. It is recorded in annual intervals using the YYYY format.
- values: the variable value is the annual salary in both euro and the national currency

Variables included in GPG_DATA

- freq: This variable shows how often the data is collected over time. In this dataset, it is recorded annually (A).
- unit: shows the measurement scale used for the data values. The unit of measures used in this dataset is percentage (PC).
- nace_r2: The variable nace_r2 identifies the economic sector according to the NACE Rev. 2 classification. The category B-S excluding O includes industry, construction, and service activities, while excluding public administration, defence, and compulsory social security.
- geo: The variable geological entity 41 territorial units. It includes 37 individual European countries, represented by ISO 3166-1 alpha-2 codes, as well as four supranational aggregates.

- TIME_PERIOD: The time variable shows the time period covered by the dataset, which spans from 2007 to 2024. It is recorded in annual intervals using the YYYY format.
- values: The variable value represents the unadjusted Gender Pay Gap (GPG), which is the percentage difference in average gross hourly earnings between male and female employees, relative to male earnings.

Part 3 - Quantifying

3.1 Data cleaning

In this section, we clean the Gender Pay Gap and salary datasets to prepare them for analysis. First, we check for missing values and remove incomplete observations, since missing data could affect the reliability of our results. Then, we keep only the variables needed for the analysis: country, year, and values.For the salary dataset, we keep only observations measured in euros (EUR) so that the salary values are comparable. We also remove aggregate regions such as EA19, EA20, and EU27_2020 because our analysis focuses on individual European countries. Finally, we rename the value variables and merge the two datasets by country and year.

3.2 Generate necessary variables

Here are descriptions for the three variables created in section 3.2:

Variable 1 – `region`
This categorical variable assigns each country to one of four European regions: Northern, Western, Southern, or Eastern Europe. The classification follows the United Nations geoscheme and is intended to uncover broad geographical patterns in the gender pay gap. By grouping countries into regions, we can compare average pay gaps and salary levels across culturally and economically similar blocks, rather than analysing individual countries in isolation.

Variable 2 – `gpg_severity`
This continuous variable translates the relative gender pay gap (expressed as a percentage) into an absolute annual earnings shortfall in euros. It is calculated as
`gpg_severity = (gpg / 100) * salary`, where `gpg` is the unadjusted gender pay gap in percent and `salary` is the average full-time adjusted salary per employee (in EUR). The resulting value represents the approximate amount, in euros, by which women's average annual earnings lag behind men's. While the percentage gap indicates the relative inequality, this monetary severity measure highlights the real-world economic impact on women in different countries and years.

Variable 3 – `region_summary` (aggregated dataset)
This is not a single variable but a summary table created by aggregating the merged country-level data. For each region and year, it computes the mean gender pay gap (`mean_gpg`), the mean salary (`mean_salary`), the mean GPG severity (`mean_gpg_severity`), and the number of countries included (`n_countries`). These regional aggregates are used to produce the temporal and spatial visualisations in the report, smoothing out country-specific noise and revealing broader regional trends over time.

Variable 1

Variable 2

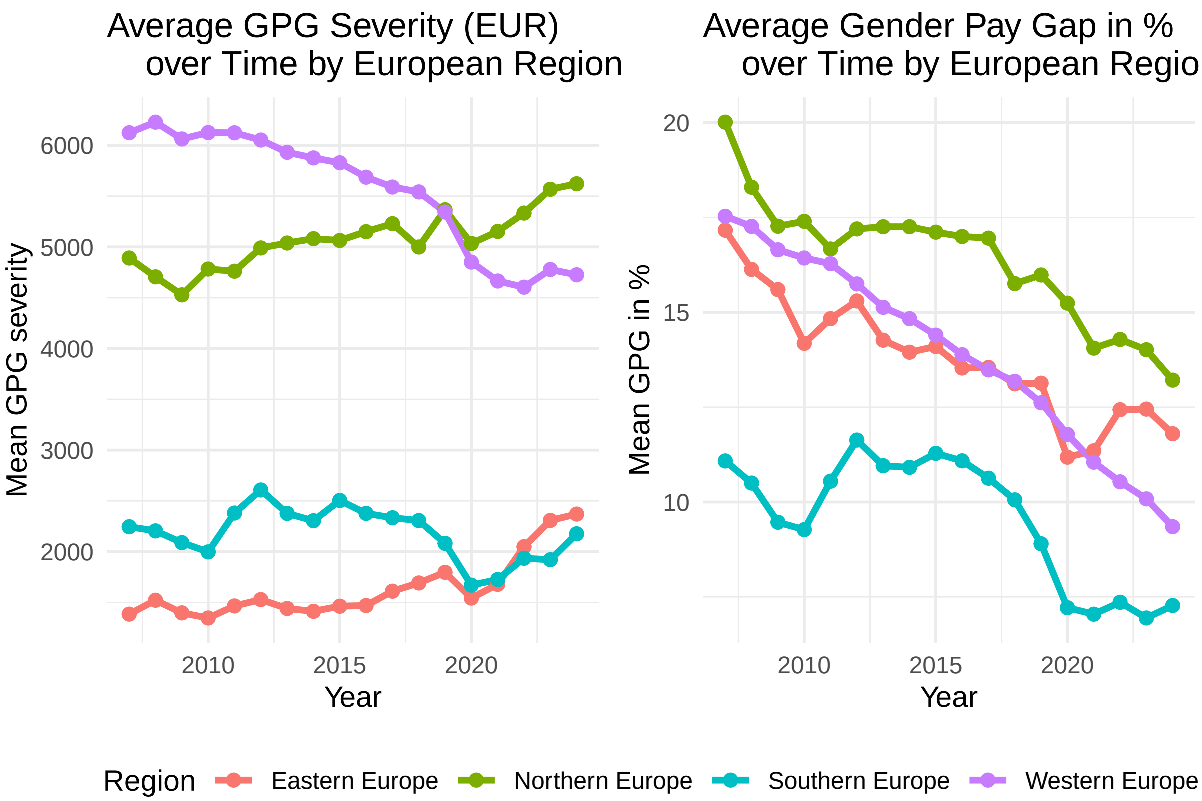
Variable 3

3.3 Visualize temporal variation

This plot shows how the Gender Pay Gap differs across European regions in the most recent year of the dataset. The boxplots allow us to compare both the average level and the spread of the Gender Pay Gap within each region. The diamond markers show the regional means, making it easier to identify which regions have higher or lower average gaps.

The results suggest that Northern Europe has the highest average Gender Pay Gap, followed by Eastern Europe and Western Europe, while Southern Europe has the lowest average gap. However, the spread also differs across regions. Western Europe and Eastern Europe show more variation between countries, while Southern Europe appears to have a smaller range of values.

This is relevant to our social problem because it shows that gender pay inequality is not evenly distributed across Europe. Regional differences suggest that economic structures, labour market policies, and social norms may influence the size of the Gender Pay Gap.



3.4 Visualize spatial variation

This plot shows how the Gender Pay Gap differs across European regions in the most recent year of the dataset. The boxplots allow us to compare both the average level and the spread of the Gender Pay Gap within each region. The diamond markers show the regional means, making it easier to identify which regions have higher or lower average gaps.

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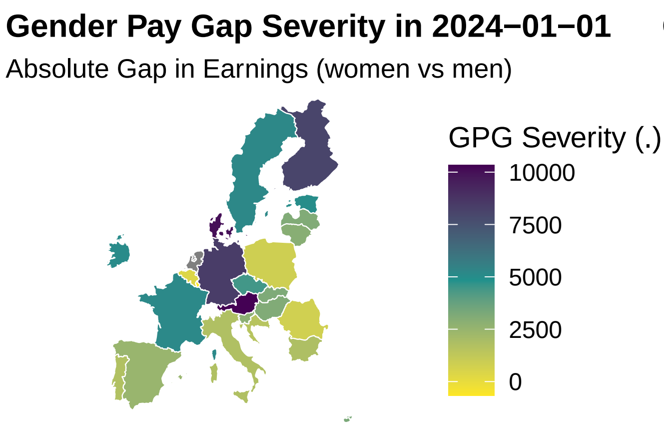

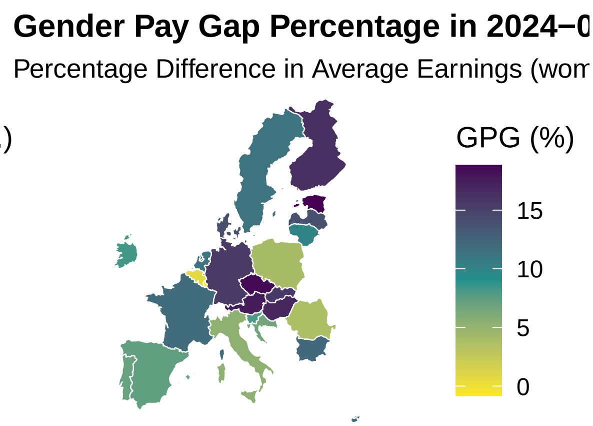

Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

This plot shows how the Gender Pay Gap differs across European regions in the most recent year of the dataset. The boxplots allow us to compare both the average level and the spread of the Gender Pay Gap within each region. The diamond markers show the regional means, making it easier to identify which regions have higher or lower average gaps.

The results suggest that Northern Europe has the highest average Gender Pay Gap, followed by Eastern Europe and Western Europe, while Southern Europe has the lowest average gap. However, the spread also differs across regions. Western Europe and Eastern Europe show more variation between countries, while Southern Europe appears to have a smaller range of values.

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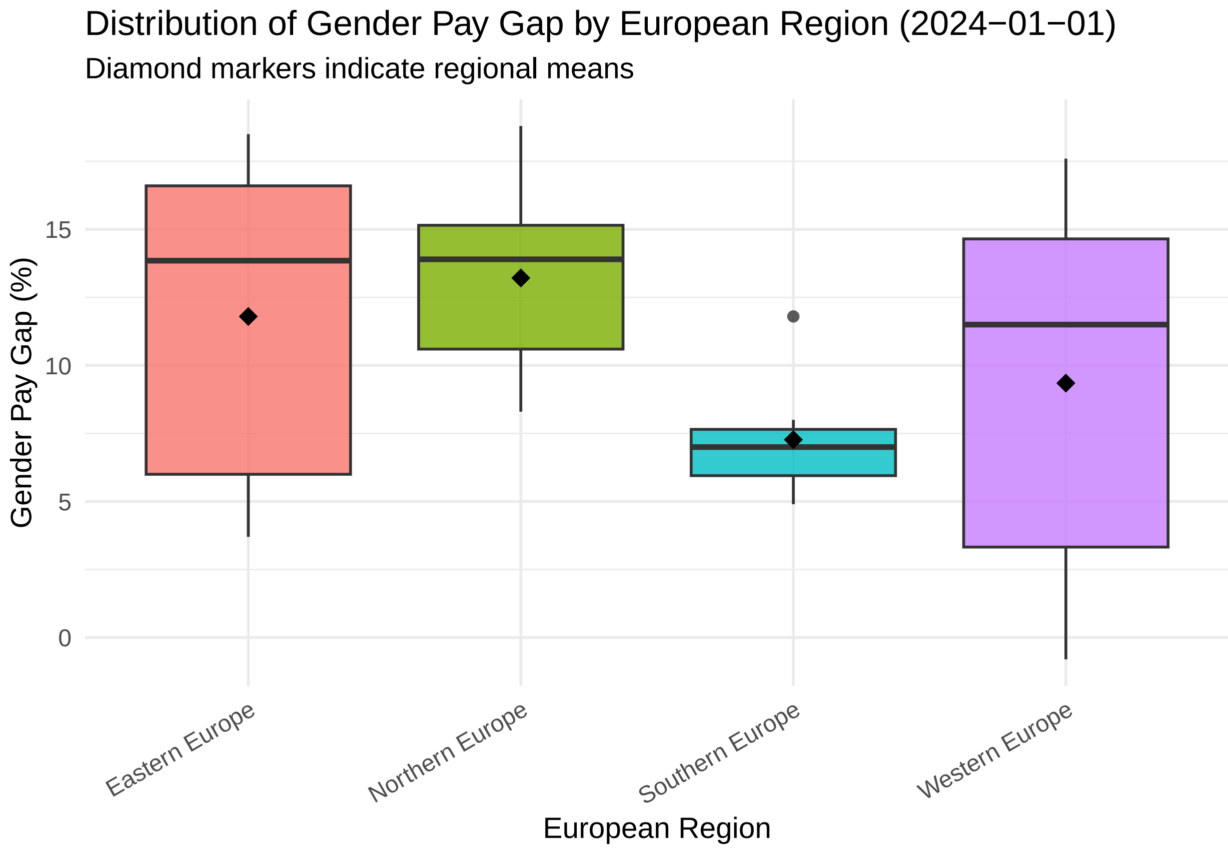


Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

This plot shows how the Gender Pay Gap differs across European regions in the most recent year of the dataset. The boxplots allow us to compare both the average level and the spread of the Gender Pay Gap within each region. The diamond markers show the regional means, making it easier to identify which regions have higher or lower average gaps.

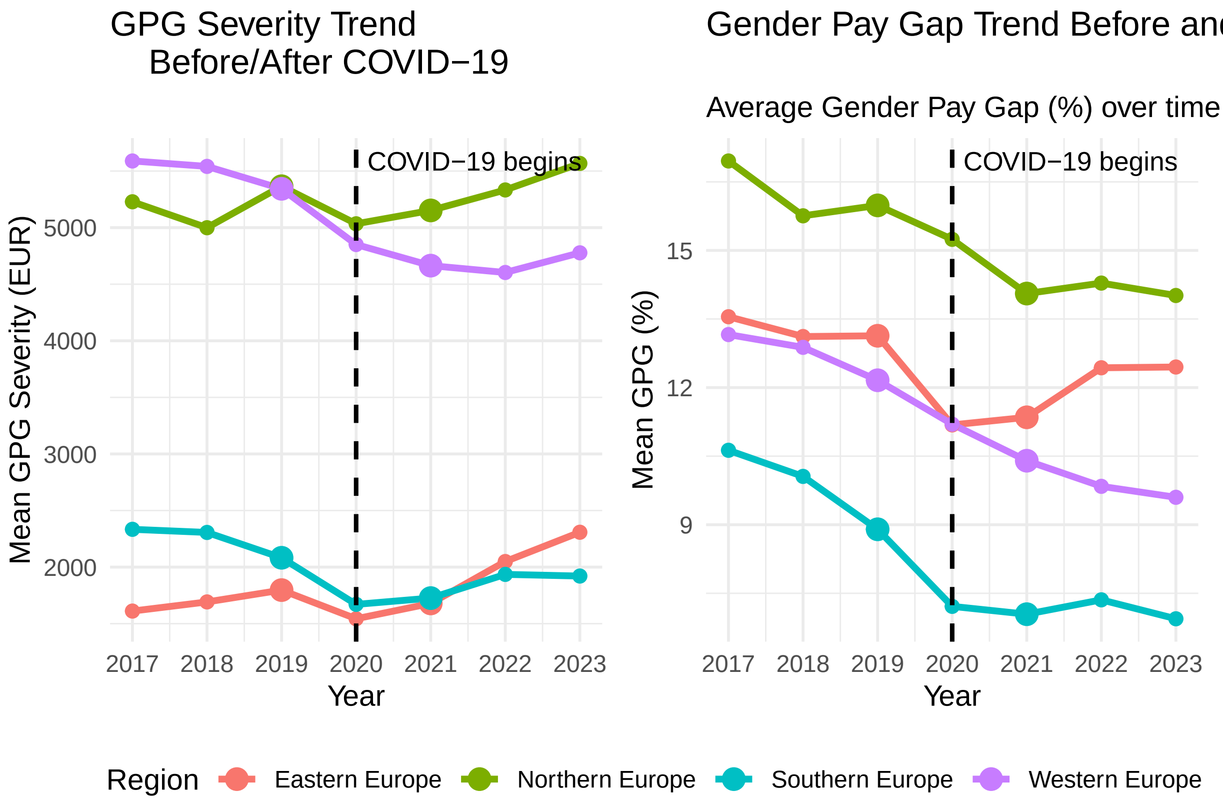
The results suggest that Northern Europe has the highest average Gender Pay Gap, followed by Eastern Europe and Western Europe, while Southern Europe has the lowest average gap. However, the spread also differs across regions. Western Europe and Eastern Europe show more variation between countries, while Southern Europe appears to have a smaller range of values.

This is relevant to our social problem because it shows that gender pay inequality is not evenly distributed across Europe. Regional differences suggest that economic structures, labour market policies, and social norms may influence the size of the Gender Pay Gap.

For the event analysis, we use COVID-19 as the relevant event. COVID-19 began in 2020 and strongly affected European labour markets through lockdowns, remote work, reduced working hours, job losses, and increased unpaid care responsibilities. These changes may have affected men and women differently, making COVID-19 relevant for studying the Gender Pay Gap.

This section uses two visualizations. The first figure shows the trend in the Gender Pay Gap around the COVID-19 period. The second figure focuses specifically on the comparison between 2019 and 2021, with 2019 representing the pre-COVID period and 2021 representing the post-COVID period.

```
## Dataset query already saved in cache_list.json...

## Reading cache file /var/folders/zv/qz_pz5hn1wv9rzx1z1nxp_n40000gn/T//Rtmpn5TxGN/eurostat/151f6a39c2e8

## Table  tesem180  read from cache file:  /var/folders/zv/qz_pz5hn1wv9rzx1z1nxp_n40000gn/T//Rtmpn5TxGN/

## Dataset query already saved in cache_list.json...

## Reading cache file /var/folders/zv/qz_pz5hn1wv9rzx1z1nxp_n40000gn/T//Rtmpn5TxGN/eurostat/a09df99951c

## Table  nama_10_fte  read from cache file:  /var/folders/zv/qz_pz5hn1wv9rzx1z1nxp_n40000gn/T//Rtmpn5T
```



Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

This study examined regional differences in the gender pay gap across Europe using Eurostat data on average full-time adjusted salaries and the unadjusted gender pay gap. The objective was to determine whether meaningful differences exist between European regions and to explore how these differences evolved over time. The results indicate that gender-based earnings inequalities persist throughout Europe, although their magnitude varies substantially between regions. While Western and Northern Europe generally exhibit higher salary levels, they also experience some of the highest levels of gender pay gap severity when measured in monetary terms. In contrast, Southern Europe displays relatively lower average gender pay gaps and lower associated monetary losses. Furthermore, the temporal analysis suggests that the gender pay gap has generally declined across Europe over the observed period. However, despite this improvement, no region has completely eliminated wage disparities between men and women. These findings suggest that economic development and rising salary levels alone are insufficient to fully close gender-based earnings differences. Overall, the study demonstrates that gender wage inequality remains a persistent feature of European labour markets and that the economic consequences of the gap depend not only on the percentage difference in earnings but also on underlying salary levels.

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: . . .

5.2 Reference list

Use APA referencing throughout your document.

include=FALSE -> to be used when we want to hide code